

Genitourinary tract infections and SGLT2 inhibitors in heart failure: an EMPEROR-Pooled analysis

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Abstract

Aims

Sodium glucose co-transporter-2 inhibitors (SGLT2i) improve outcomes of patients with heart failure (HF). Urinary tract infections (UTI) are relatively common in HF, and their occurrence has been associated with an increased mortality risk. The use of SGLT2i has been associated with the occurrence of UTIs and mycotic genitourinary infections (MGI) which are a major cause of treatment discontinuation. This study aimed to study genitourinary infections in men and women with HF, their association with outcomes, and the impact of empagliflozin.

Methods

Analysis of EMPEROR-Pooled including a total of 9718 patients across the spectrum of ejection fraction. Descriptive statistics, Cox and time-varying survival models were used. The median follow-up time was 21 months.

Results

Throughout follow-up, 757 (7.8%) patients experienced a lower UTI, 60 (0.6%) MGI, and 41 pyelonephritis/urosepsis (0.4%). Genitourinary infections were more frequent in women than in men. Compared to patients without UTI/MGI, those with pyelonephritis/urosepsis were older and had higher comorbidity burden, whereas those with MGI were younger and had a higher prevalence of diabetes and obesity. Compared to placebo, empagliflozin was associated with a higher risk of lower UTI and MGI, particularly balanitis in men, but not pyelonephritis/urosepsis. The risk of death increased after UTI, but not after MGI. The effect of empagliflozin versus placebo on the composite of HF hospitalization or cardiovascular death was not modified by the occurrence of genitourinary infections.

Conclusions

Lower UTIs were relatively frequent in HF patients, whereas MGI and pyelonephritis/urosepsis were infrequent. SGLT2i increased the risk of lower UTI and MGI. Although UTI, but not MGI, were associated with a higher subsequent risk of death, the clinical benefits of SGLT2i were preserved regardless of genitourinary infection occurrence.

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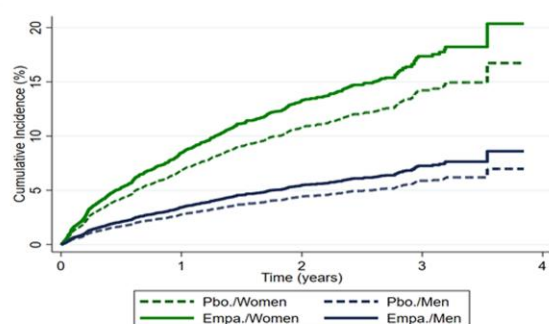
Graphical abstract

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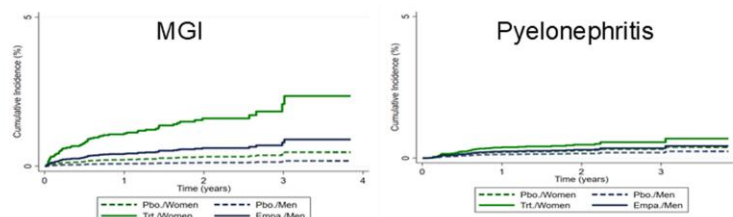
SGLT2 inhibitors improve HF outcomes, but their use has been associated with the occurrence of urinary tract infections (UTIs) and mycotic genitourinary infections (MGI) which are a cause of treatment discontinuation

Analysis of EMPEROR-Pooled including a total of 9718 patients across the spectrum of ejection fraction to study UTIs and MGIs and their association with empagliflozin treatment

Lower UTIs were more frequent in women than men and empagliflozin increased their risk by 1.2-fold vs placebo



MGIs and pyelonephritis were relatively uncommon; still, empagliflozin was associated with an increased risk of MGIs in women and balanitis among men



Empagliflozin reduced HF hospitalizations and cardiovascular death irrespective of the occurrence of UTIs or MGIs

While empagliflozin increased the risk of urinary tract infections and mycotic genital infections, its benefits were maintained irrespective of the occurrence of such infections.

Keywords

Genitourinary infections • Sodium glucose co-transporter 2 inhibitors • Heart failure

Introduction

Sodium glucose co-transporter 2 inhibitors (SGLT2i) improve outcomes in patients with heart failure (HF) across the full spectrum of ejection fraction, with benefits that persist while therapy is continued.^{1–3}

Urinary tract infections (UTI) are relatively common in HF, particularly among elderly patients with co-morbidities such as chronic kidney disease and type 2 diabetes (T2D).⁴ UTIs have been associated with a higher subsequent risk of cardiovascular and kidney events.⁵ Because SGLT2i therapy has been associated with genitourinary infections, including UTIs and mycotic genital infections (MGI), concerns about infections remain a major cause of SGLT2i discontinuation, which may further contribute to adverse outcomes among people who develop genitourinary infections.⁶

In HF, the incidence and prognostic implications of genitourinary infections, and their relationship with SGLT2i treatment, require further characterization.⁷ In particular, data are limited on infection subtypes (lower UTI/cystitis, MGI, pyelonephritis/urosepsis), the impact of SGLT2i on each subtype, and sex-specific differences in infection risk. To address these gaps, we analysed the EMPEROR-Program to provide a comprehensive assessment of genitourinary infections and SGLT2i-associated risk in HF across the full range of left ventricular ejection fraction (LVEF)⁸

Methods

Study design and patient population

The design and primary results of EMPEROR-Pooled have been previously published.^{8,9} In brief, the EMPEROR-Pooled combined individual patient

data from EMPEROR-Reduced (EMPagliflozin outcome trial in Patients With chrOnic heaRt Failure With Reduced Ejection Fraction; NCT03057977) and EMPEROR-Preserved (EMPagliflozin outcome trial in Patients With chrOnic heaRt Failure With Preserved Ejection Fraction; NCT03057951), the two phase III international, multicentre, randomized, double-blind, parallel-group, placebo-controlled trials that enrolled adult patients with chronic HF with New York Heart Association (NYHA) class II–IV symptoms for at least 3 months and elevated natriuretic peptide levels across a wide range of LVEF ($\leq 40\%$ in EMPEROR-Reduced and $>40\%$ in EMPEROR-Preserved).^{1,2} The protocol of each trial complied with the Declaration of Helsinki, was approved by the Ethical Committee of the participating sites, and all patients gave written informed consent to participate in the study.

The median follow-up time was 16 months in EMPEROR-Reduced, 26 months in EMPEROR-Preserved, and 21 (15–28) months in EMPEROR-Pooled.

Randomization

Patients were randomized in a double-blind manner (in a 1:1 ratio) to receive placebo or empagliflozin 10 mg daily, in addition to their usual therapy. Following entry into the trial, all appropriate treatments for HF or other medical conditions could be initiated, discontinued, or altered at the clinical discretion of the investigator.

Endpoints

The primary endpoint in EMPEROR-Pooled was a composite of adjudicated cardiovascular death or hospitalization for HF, analysed as the time-to-first event.

The outcome of interest in the present analysis was the occurrence of genitourinary infections, captured as investigator-reported diagnoses and

Table 1 Baseline characteristics of the patients by type of genitourinary infection

Characteristics	None	Lower UTI	MGI	Pyelon./uroseps.	P
N.	8860	757	60	41	
Age, years	69.7 (10.4)	72.9 (10.1)	66.4 (11.4)	75.1 (8.2)	<.001
Women	3045 (34.4%)	467 (61.7%)	36 (60%)	21 (51.2%)	<.001
Hypertension	7352 (83.0%)	680 (89.8%)	53 (88.3%)	37 (90.2%)	<.001
T2D	4262 (48.1%)	401 (53.0%)	37 (61.7%)	26 (63.4%)	.002
AFib/flutter	4130 (46.6%)	404 (53.4%)	27 (45.0%)	24 (58.5%)	.002
HFH prior 12mo	2558 (28.9%)	195 (25.8%)	16 (26.7%)	19 (46.3%)	.021
Ischemic HF	3739 (42.2%)	269 (35.5%)	26 (43.3%)	12 (29.3%)	.001
NYHA II	7064 (79.7%)	548 (72.4%)	46 (76.7%)	29 (70.7%)	<.001
SBP, mmHg	127.9 (16.3)	129.5 (16.7)	129.6 (17.7)	132.2 (17.4)	.018
Heart rate, bpm	70.7 (11.8)	70.7 (11.7)	71.2 (11.2)	70.2 (16.8)	.98
Per. oedema	2299 (25.9%)	271 (35.8%)	22 (36.7%)	8 (19.5%)	<.001
Pulm. rales	715 (8.1%)	69 (9.1%)	5 (8.3%)	2 (4.9%)	.66
BMI, kg/m ²	29.0 (5.7)	30.2 (6.0)	32.0 (6.6)	29.1 (6.0)	<.001
LVEF, %	43.6 (15.1)	48.8 (14.8)	47.4 (14.4)	53.1 (14.5)	<.001
Hgb, g/dL	13.5 (1.6)	13.0 (1.6)	13.4 (1.4)	12.9 (1.6)	<.001
UACR, mg/g	21 (8, 73)	28 (10, 93)	22 (8, 100)	37 (18, 162)	<.001
eGFR, mL min	61.6 (20.5)	56.7 (20.3)	63.8 (19.8)	47.6 (21.3)	<.001
NTproBNP, pg/mL	1295 (675, 2334)	1256 (650, 2321)	1007 (568, 1845)	1871 (953, 2543)	.14
Loop diur.	6570 (74.2%)	560 (74.0%)	43 (71.7%)	31 (75.6%)	.97
ACEi/ARB	6681 (75.4%)	552 (72.9%)	44 (73.3%)	28 (68.3%)	.33
Beta-blocker	7940 (89.6%)	669 (88.4%)	57 (95.0%)	34 (82.9%)	.17
MRA	4527 (51.1%)	337 (44.5%)	25 (41.7%)	16 (39.0%)	.001
Thiazide	1381 (15.6%)	124 (16.4%)	10 (16.7%)	9 (22.0%)	.66
EMP.-reduced	3515 (39.7%)	189 (25.0%)	17 (28.3%)	9 (22.0%)	<.001
EMP.-preserved	5345 (60.3%)	568 (75.0%)	43 (71.7%)	32 (78.0%)	

UTI, urinary tract infection; MGI, mycotic genitourinary infections; T2D, type 2 diabetes; HFH, heart failure hospitalization; NYHA, New York Heart Association; SBP, systolic blood pressure; BMI, body mass index; LVEF, left ventricular ejection fraction; UACR, urinary albumin-to-creatinine ratio; eGFR, estimated glomerular filtration rate; ACEi/ARB, angiotensin converting enzyme inhibitor/angiotensin receptor blocker; MRA, mineralocorticoid receptor antagonist.

recorded as adverse events or serious adverse events (i.e. adverse events of special interest) using Medical Dictionary for Regulatory Activities (MedDRA) definitions.¹⁰ Analyses were performed through the end of planned treatment plus 7 days, consistent with the safety analyses of the EMPEROR-Program.

Statistical analysis

Baseline characteristics were compared between patients who developed genitourinary infections, specifically lower UTI, MGI, and pyelonephritis/urosepsis and those who did not, using the appropriate descriptive statistics and tests. Multinomial regression modelling with stepwise selection was used to test the variables with the strongest association with each genitourinary infection subtype. Competing risks models¹¹ were used to study the association of randomized treatment (empagliflozin vs placebo) with the occurrence of genitourinary infections. All-cause mortality was defined as 'competing risk', and all models were adjusted for the all variables with a *P*-value <.05 in Table 1. The association between empagliflozin treatment and total genitourinary infections was studied with negative binomial regression models accounting for follow-up time. The association between the occurrence of genitourinary infections and subsequent HF hospitalization or cardiovascular death, and all-cause death was assessed using time-updated model with follow-up time set to 0, in both the treatment and placebo groups, at the time of genitourinary infection occurrence. To explore whether the occurrence of genitourinary infections modified the effect of empagliflozin on the study primary outcome, a treatment-by-infection interaction term was inserted in the model while adjusting for the all variables with a *P*-value <.05 in Table 1.

Adverse events (AEs) were retrieved from text descriptions in the case report form (CRF) following the MedDRA. *P*-values and 95% confidence intervals presented in this report have not been adjusted for multiplicity, given the exploratory nature of this work. All analyses were performed using Stata®, version 18 (StataCorp. 2023. Stata Statistical Software: Release 18. College Station, TX: StataCorp LLC). All analysis here presented were performed within the Vivli® (<https://vivli.org/>) platform, and the trial sponsor was not involved in this analysis.

Results

Patients' characteristics by UTI subtype

A total of 9718 patients were included. During follow-up, 757 (7.8%) patients experienced a lower UTI, 60 (0.6%) a MGI, and 41 pyelonephritis or urosepsis (0.4%). Median (pct₂₅₋₇₅) time to first event was 8.2 (2.8–15.6) months for lower UTI, 5.8 (1.5–14.0) months for MGI, and 6.8 (2.8–10.7) months for pyelonephritis or urosepsis. Compared with patients who did not experience UTI or MGI, those with pyelonephritis or urosepsis were the oldest, had a near balanced sex distribution, and had the highest comorbidity burden, with more frequent T2D, atrial fibrillation or flutter (AFib), and prior HF hospitalization, lower haemoglobin and eGFR, higher UACR, and higher LVEF. Patients with MGI were the youngest and were more often women, with a high prevalence of T2D and the highest body mass index

Table 2 Association of empagliflozin treatment and the occurrence of genitourinary infection by sex, accounting for the competing risks of death

UTI type	n/N (%) events ^a		Empa. vs Pbo. Adj. sHR (95%CI)	P-value Int.-P
	Pbo.	Empa.		
Lower UTI				
Overall	343/3894 (8.8) IR = 4.1 (3.7–4.6)	414/3858 (10.7) IR = 5.1 (4.6–5.6)	1.24 (1.08–1.43)	0.003
Women	214/1394 (15.4) IR = 6.8 (6.0–7.8)	253/1362 (18.6) IR = 8.3 (7.4–9.4)	1.24 (1.03–1.49)	0.91
Men	129/2500 (5.2) IR = 2.5 (2.1–2.9)	161/2496 (6.5) IR = 3.1 (2.7–3.6)	1.24 (0.98–1.57)	
Mycotic genitourinary infection				
Overall	10/4158 (0.2) IR = 0.1 (0.1–0.2)	50/4142 (1.2) IR = 0.6 (0.5–0.8)	5.07 (2.57–10.0)	<0.001
Women	9/1567 (0.6) IR = 0.3 (0.1–0.6)	27/1548 (1.7) IR = 0.9 (0.6–1.3)	2.99 (1.40–6.38)	0.067
Men	1/2591 (<0.1) IR = 0.0 (0.0–0.1)	23/2594 (0.9) IR = 0.4 (0.3–0.7)	22.0 (2.94–165)	
Pyelonephritis/urosepsis				
Overall	15/4157 (0.4) IR = 0.2 (0.1–0.3)	26/4170 (0.6) IR = 0.3 (0.2–0.5)	1.80 (0.95–3.43)	0.073
Women	4/1572 (0.3) IR = 0.1 (0.0–0.3)	17/1561 (1.1) IR = 0.6 (0.3–0.9)	4.55 (1.54–13.5)	0.019
Men	11/2585 (0.4) IR = 0.2 (0.1–0.4)	9/2609 (0.3) IR = 0.2 (0.1–0.3)	0.74 (0.32–1.69)	

UTI, urinary tract infection; IR, incidence rate per 100 person-years; Adj., models adjusted for all variables with a $P < .05$ in Table 1.

Median time-to-lower UTI = 250 (85–475) days; median time-to-MGI = 177 (47–428) days; median time-to-pyelonephritis/urosepsis = 206 (84–325) days.

^aCompeting-risk events excluded from the total.

(BMI). Patients with lower UTI exhibited an ‘intermediate’ phenotype, younger with fewer comorbidities than those with pyelonephritis urosepsis, but older than those with MGI (Table 1). Factors independently associated with each genitourinary infection subtype are displayed in Supplementary Table S1. Lower UTIs were more frequent in EMPEROR-Preserved than in EMPEROR-Reduced, likely reflecting the older population enrolled in EMPEROR-Preserved.

Empagliflozin and the occurrence of genitourinary infections by sex

Compared with placebo, empagliflozin treatment was associated with a higher risk of lower UTI (placebo 4.1 events/100py vs empagliflozin 5.1 events/100py, adjusted HR 1.24, 95%CI 1.08–1.43, $P = .003$). Lower UTI incidence was 2.5 to 3-fold higher in women than in men, but the relative risk associated with empagliflozin treatment was similar in both sexes (interaction- $P = .91$). Despite a low absolute risk, empagliflozin markedly increased the risk of MGI (placebo 0.1 events/100py vs empagliflozin 0.6 events/100py, adjusted HR 5.07, 95% CI 2.57–10.0, $P < .001$). The relative increase in risk appeared greater in men (predominantly balanitis) than in women (interaction- $P = .067$). Empagliflozin was not significantly associated with pyelonephritis or urosepsis (placebo 0.2 events/100py vs empagliflozin 0.3 events/100py, adjusted HR 1.80, 95%CI 0.95–3.43, $P = .073$). Nonetheless, the empagliflozin-associated risk appeared higher in women than in men (interaction- $P = .019$; Table 2 and Figure 1).

MedDRA preferred terms by sex for each infection subtype are shown in Supplementary Table S2. Analyses of total (recurrent) genitourinary infections were consistent with the first event analyses. It should be noted

that 24% of patients on empagliflozin and 17% on placebo experienced a recurrent uncomplicated UTI with an average time between the first and second events of 4 months. Recurrent MGI and pyelonephritis or urosepsis were uncommon, Supplementary Table S3 (including Legend).

Risk of heart failure events and mortality after a genitourinary infection

Compared with follow-up time without a lower UTI (including patients who never experience one), the risk of HF hospitalization or cardiovascular death was higher after a first lower UTI (10.5 vs 14.2 events/100py, adjusted HR 1.56, 95%CI 1.26–1.92, $P < .001$). This association was observed in both men and women, with no evidence of effect modification by sex (interaction- $P = .67$). All-cause mortality risk was also higher after a lower UTI in the overall population and within each sex (interaction- $P = .10$). In contrast, MGI was not associated with subsequent risk of HF hospitalization or cardiovascular death, nor with all-cause mortality ($P = .86$). The risk of HF hospitalization or cardiovascular death was not significantly increased after a pyelonephritis/urosepsis; however, the risk of all-cause mortality was increased: adjusted HR 3.29, 95% CI 1.87–5.76, $P < .001$, in both men and women; interaction- $P = .77$ (Table 3 and Figure 2).

Treatment effect on the study primary outcome according to genitourinary infection

The effect of empagliflozin vs placebo to reduce the incidence of the primary composite outcome of HF hospitalization or cardiovascular

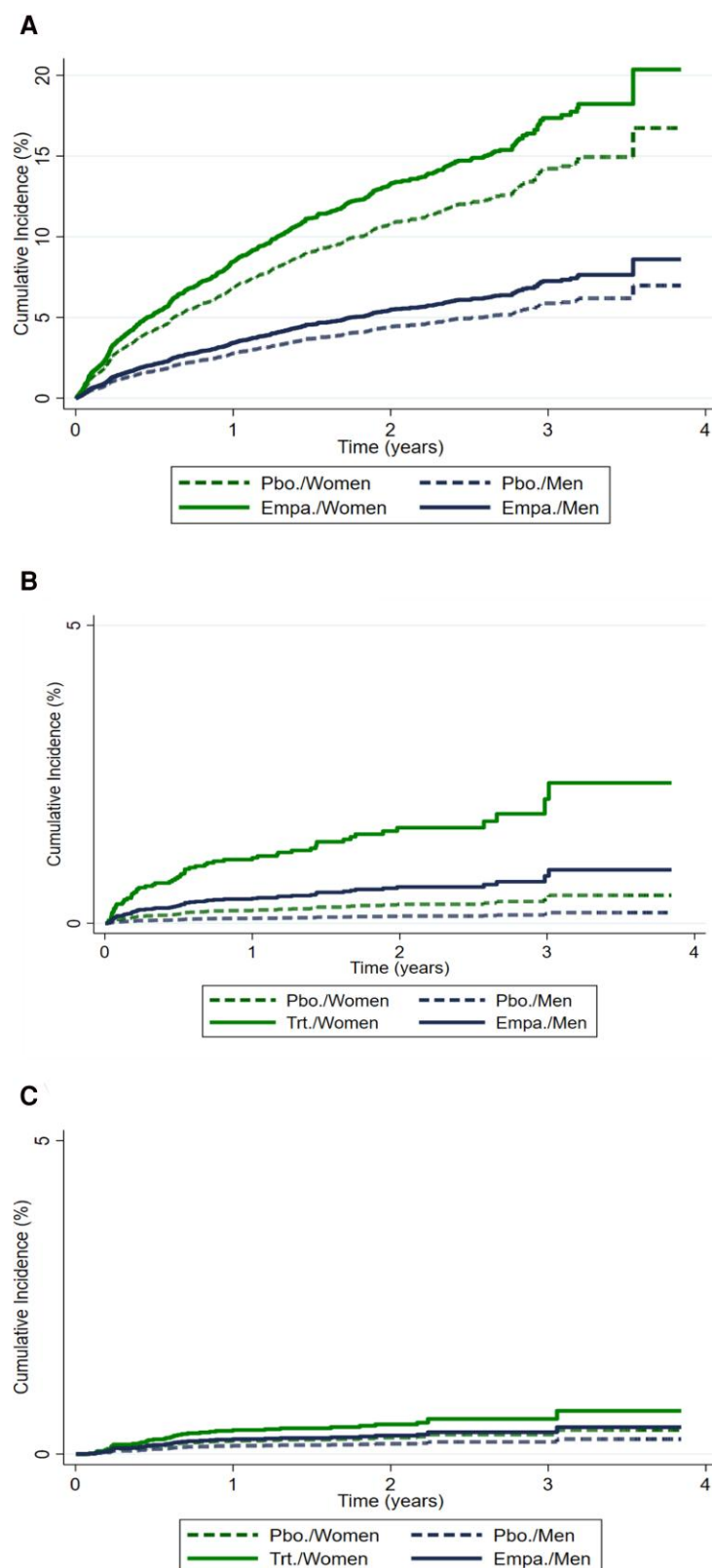


Figure 1 Occurrence of genitourinary infections by treatment and sex. (A) Lower UTI. (B) Mycotic genitourinary infection. (C) Pyelonephritis/ur-osepsis. UTI, urinary tract infection; Pbo., placebo; Empa., empagliflozin.

Table 3 Risk HF hospitalization and mortality after a genitourinary infection

Event/outcome/ subgroup	n/N (%) events before/without UTI IR per 100py (95%CI)	n/N (%) events after UTI IR per 100py (95%CI)	UTI vs no UTI Adj. HR (95%CI)	UTI-by-sex int. P
Lower UTI				
CVD or HFH				
Overall	1645/9069 (18.1) 10.5 (10.0–11.0)	104/649 (16.0) 14.2 (11.7–17.2)	1.56 (1.26–1.92) P < .001	.67
Women	490/3162 (15.5) 8.2 (7.5–9.0)	62/407 (15.2) 12.8 (10.0–16.5)	1.57 (1.19–2.07)	
Men	1155/5907 (19.6) 11.9 (11.2–12.6)	42/242 (17.4) 16.9 (12.5–22.9)	1.56 (1.12–2.17)	
ACM				
Overall	1209/8976 (13.5) 7.2 (6.9–7.7)	155/742 (20.9) 18.3 (15.7–21.5)	2.31 (1.92–2.78) P < .001	.10
Women	346/3113 (11.1) 5.5 (5.0–6.2)	75/456 (16.5) 13.5 (10.8–17.0)	2.00 (1.53–2.62)	
Men	863/5863 (14.7) 8.3 (7.7–8.8)	80/286 (28.0) 27.4 (2.0–34.2)	2.66 (2.06–3.45)	
Mycotic genitourinary infection				
CVD or HFH				
Overall	1743/9661 (18.0) 10.7 (10.2–11.2)	6/57 (10.5) 7.8 (3.5–17.4)	0.93 (0.42–2.06) P = .86	NC
Women	550/2985 (15.6) 8.6 (7.9–9.4)	2/34 (5.9) 4.0 (1.0–15.9)	0.55 (0.13–2.33)	
Men	1193/6126 (19.5) 12.0 (11.3–12.7)	4/23 (17.4) 15.1 (5.7–40.2)	1.45 (0.59–3.59)	
ACM				
Overall	1358/9658 (14.1) 7.8 (7.4–8.2)	6/60 (10.0) 7.1 (3.2–15.8)	0.90 (0.41–1.99) P = .89	NC
Women	418/3533 (11.8) 6.2 (5.6–6.8)	3/36 (8.3) 5.5 (1.8–17.0)	0.86 (0.27–2.75)	
Men	940/6125 (15.4) 8.8 (8.3–9.4)	3/24 (12.5) 10.0 (3.2–30.9)	0.89 (0.31–2.59)	
Pyelonephritis/urosepsis				
CVD or HFH				
Overall	1745/9685 (18.0) 10.6 (10.2–11.2)	4/33 (12.1) 11.5 (4.3–30.5)	1.13 (0.42–3.00) P = .81	NC
Women	550/3550 (15.5) 8.6 (7.9–9.3)	2/19 (10.5) 8.4 (2.1–33.4)	0.97 (0.24–3.83)	
Men	1195/6135 (19.5) 12.0 (11.3–12.7)	2/14 (14.3) 18.2 (4.6–72.9)	1.60 (0.39–6.50)	
ACM				
Overall	1351/9678 (14.0) 7.7 (7.3–8.2)	13/40 (32.5) 31.9 (18.5–54.9)	3.29 (1.87–5.76) P < .001	.77
Women	416/3549 (11.7) 6.1 (5.6–6.8)	5/20 (25.0) 19.1 (7.9–45.8)	2.39 (0.97–5.86)	
Men	935/6129 (15.3) 8.7 (8.2–9.3)	8/20 (40.0) 54.9 (27.4–109)	4.44 (2.13–9.25)	

UTI, urinary tract infection; CVD, cardiovascular death; HFH, heart failure hospitalization; ACM, all-cause mortality; int., interaction.

NC, not computed due to small number ($n < 5$) of events post-UTI.Adj., models adjusted for all variables with a $P < .05$ in Table 1 plus treatment allocation.

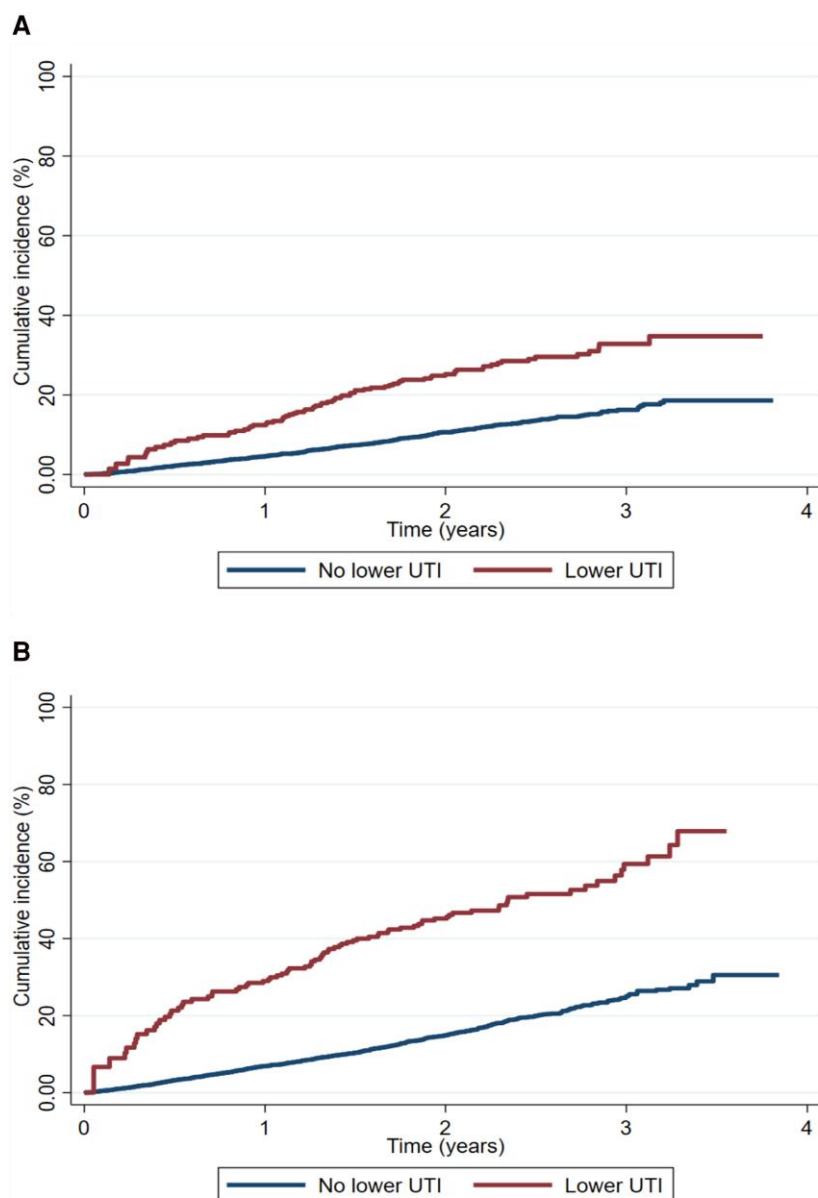


Figure 2 Risk of mortality after a genitourinary infection in men and women. (A) Women, lower UTI. (B) Men, lower UTI. (C) Women, pyelonephritis/urosepsis. (D) Men, pyelonephritis/urosepsis. UTI, urinary tract infection.

death was not modified by the occurrence of lower UTI, MGI, or pyelonephritis/urosepsis; interaction- $P > .1$ for all genitourinary infection subtypes (Table 4).

Adverse event severity and treatment discontinuation by genitourinary infection

Fewer than 10% of lower UTIs in women and fewer than 20% in men were classified as serious adverse events (SAEs). No MGI were reported as SAE, whereas most pyelonephritis or urosepsis events were classified as SAEs. Investigators attributed causality to study treatment in 30–40% of lower UTIs in both sexes and in more than 50% of MGI in both sexes. Treatment interruption or discontinuation occurred in more than 20% of patients after a pyelonephritis or urosepsis and

was less frequent after other infection subtypes. A total of 17 pyelonephritis events occurred after a prior lower UTI (Supplementary Table S4). After a lower UTI, treatment withdrawal or discontinuation was more frequent with empagliflozin than with placebo ($P = .018$). No statistically significant between treatment group differences were observed in treatment withdrawal or discontinuation after MGI or after pyelonephritis or urosepsis (Supplementary Table S5).

Discussion

This study provides a detailed description of genitourinary infections in more than 9700 patients with HF across the full spectrum of ejection fraction. Over a median follow-up of nearly 2 years, genitourinary

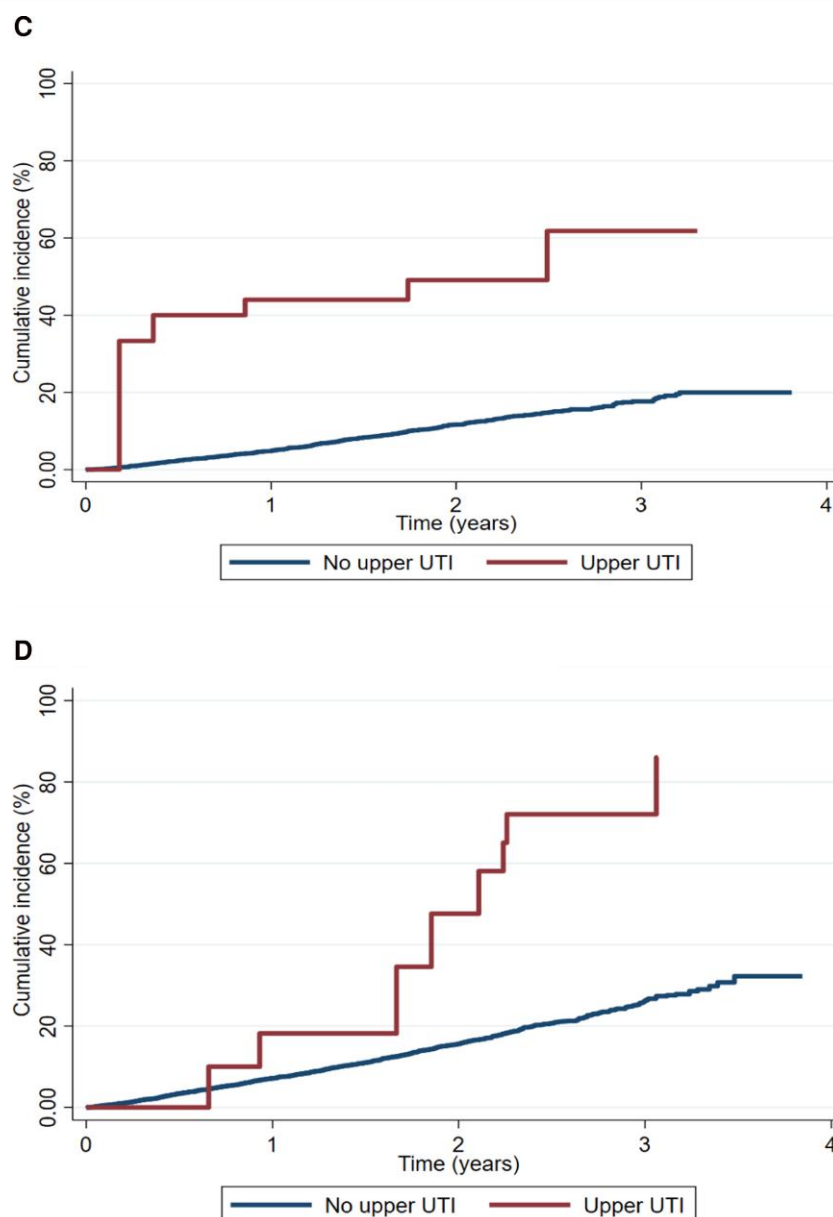


Figure 2 Continued

infections were overall relatively uncommon. Still, uncomplicated lower UTI was the most frequent subtype, occurring at a rate of 4–5% per year, typically early after randomization with a higher incidence in women. Empagliflozin modestly increased the risk of lower UTI (approximately 1.2-fold), and a first lower UTI identified a higher risk phenotype, with a 1.5- to 2-fold higher subsequent risk of cardiovascular events and mortality.

Pyelonephritis or urosepsis was higher among older patients with a higher comorbidity burden. Although these severe events carried substantial clinical consequences, including higher rates of treatment interruption and a marked increase in all-cause mortality, their absolute incidence was low (<0.5%/year) and was not significantly increased by empagliflozin in the overall population. Nonetheless, the sex interaction suggests that women may be at higher risk of pyelonephritis or urosepsis with SGLT2i.

MGIs were also infrequent (<1%/year) but were more common in patients with obesity and T2D and increased substantially with empagliflozin (≈ 5 -fold). The relative increase appeared greater in men, largely driven by balanitis, which occurred almost exclusively in those assigned to SGLT2i (≈ 22 -fold increase). However, because balanitis aetiology was not adjudicated, a mycotic cause cannot be confirmed. Importantly, MGI were not associated with subsequent cardiovascular outcomes or mortality, were generally mild, and rarely led to treatment discontinuation.

Across all genitourinary infection subtypes, the benefits of empagliflozin on the primary composite outcome of HF hospitalization or cardiovascular death maintained irrespective of infection occurrence.

The glycosuric effect of SGLT2i has been proposed to increase susceptibility to UTIs, but much of the evidence comes from T2D populations, in whom glycosuria may be more pronounced than in those

Table 4 Treatment effect on the primary outcome according to genitourinary infection

CVD/HFH UTI subtype	n/N (%) events before/without UTI IR per 100py (95%CI)		n/N (%) events after UTI IR per 100py (95%CI)		Empa. vs Pbo. HR (95%CI) before/ without UTI	Empa. vs Pbo. HR (95%CI) after UTI	Int. P
Treatment	Pbo.	Empa.	Pbo.	Empa.			
Lower UTI	920/4571 (20.1)	725/4498 (16.1)	53/287 (18.5)	51/362 (14.1)	0.77 (0.70–0.85)	0.67 (0.45–1.00)	.52
	11.8 (11.1–12.6)	9.2 (8.5–9.9)	17.2 (13.1–22.5)	12.1 (9.2–15.9)			
MGI	972/4851 (20.0)	771/4810 (16.0)	1/7 (14.3)	5/50 (10.0)	0.77 (0.70–0.84)	0.28 (0.03–3.09)	.41
	12.0 (11.3–12.8)	9.3 (8.7–10.0)	18.5 (2.6–131.1)	7.0 (2.9–16.9)			
Pyelonephritis/ urosepsis	971/4846 (20.0)	774/4839 (16.0)	2/12 (16.7)	2/21 (9.5)	0.77 (0.70–0.84)	0.53 (0.07–3.82)	.71
	12.0 (11.3–12.8)	9.3 (8.7–10.0)	16.5 (4.1–65.9)	8.8 (2.2–35.1)			

UTI, urinary tract infection; MGI, mycotic genitourinary infection; CVD, cardiovascular death; HFH, heart failure hospitalization; Empa., empagliflozin; Pbo., placebo; int., interaction. Trt. vs Pbo. treatment effect using UTI as a time-varying covariate along with a treatment-by-UTI interaction term.

without diabetes.¹² In this context, the present analysis represents the largest evaluation of SGLT2i-associated genitourinary infections in HF patients with and without T2D. Risk determinants were mostly infection subtype- and sex-dependent.¹³ Specifically, older age, kidney disease, and a higher comorbidity burden characterized patients who developed pyelonephritis or urosepsis. While in the overall population SGLT2i did not appear to meaningfully increase the risk of this severe infection subtype, women may be at higher risk when treated with an SGLT2i. In contrast, T2D and obesity identified patients at risk for MGI, and SGLT2i substantially increased its relative risk, particularly balanitis among men. Still, women experienced the highest absolute risk of MGI while receiving SGLT2i. Lower UTI rates were higher in women, and SGLT2i was associated with a small excess risk.

UTI rates in the EMPEROR programme were broadly consistent with prior reports (6–18%),^{6,14,15} and the association between incident UTI and subsequent cardiovascular events and mortality extends previous observations.^{5,6} However, treatment discontinuation after UTI in EMPEROR was considerably lower than in 'real-world' reports,⁶ suggesting that closer trial follow-up and structured management may mitigate unnecessary discontinuation. It should be noted that compared to placebo an 'excess' of only 29 patients interrupted or withdrew empagliflozin after an UTI, supporting a negligible impact of treatment discontinuation on the occurrence adverse outcomes after an uncomplicated UTI. Importantly, because empagliflozin benefits were maintained irrespective of UTI or MGI occurrence, these findings support treatment continuation whenever feasible, even in the setting of an infection.³ When interruption is needed, a temporary pause is preferable to a permanent discontinuation, particularly given that discontinuation does not appear to reduce recurrent UTI risk.⁶ In our study a recurrent uncomplicated UTI occurred in 24% of patients on empagliflozin (vs 17% on placebo) and the average time between the first and second UTI was around 4 months, suggesting that these events were not closely related. Moreover, recurrent pyelonephritis or urosepsis were very rare. Nevertheless, in routine practice, SGLT2i discontinuation after UTI remains common and has been reported to account for nearly 40% of treatment cessations.^{16,17}

Limitations

Some limitations should be acknowledged. First, this is a *post hoc* analysis of two randomized trials, and these findings may not be generalizable to patients with characteristics that were underrepresented or excluded, such as bedridden patients or those with chronic urinary catheterization. Second, UTIs and MGIs were investigator-reported

diagnoses, and detailed microbiological data were not available, including pathogen identification or antibiotic regimen.

Conclusion

Lower UTIs were relatively common in patients with HF, whereas MGI and pyelonephritis or urosepsis were uncommon. SGLT2i increased the risk of lower UTIs in both sexes, increased the risk of pyelonephritis or urosepsis in women, and increased the risk of MGI in both sexes, with the largest relative increase for balanitis in men. Treatment discontinuation occurred in about 20% of patients after a pyelonephritis/urosepsis but was uncommon following other genitourinary infection subtypes. Although UTIs, but not MGI, were associated with a higher subsequent risk of death, the cardiovascular benefits of SGLT2i were preserved regardless of whether lower UTIs, MGIs, pyelonephritis or urosepsis occurred.

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Supplementary data

Supplementary data are available at [European Journal of Heart Failure](#) online.

Declarations

Disclosure of Interest

Dr Packer reports consulting to 89bio, Abbvie, Altimmune, Alnylam, Amarin, Amgen, Ardelyx, AstraZeneca, Boehringer Ingelheim, Caladrius, Casana, CSL Behring, Cytokinetics, Imara, Lilly, Medtronic, Moderna, Novartis, Reata, Relypsa, and Salamandra; Dr Butler has received consulting income from Abbott, American Regent, Amgen, Applied Therapeutic, AstraZeneca, Bayer, Boehringer Ingelheim, Bristol Myers Squibb, Cardiac Dimension, Cardior, CVRx, Cytokinetics, Edwards, Element Science, Innolife, Impulse Dynamics, Imbria, Inventiva, Lexicon, Lilly, LivaNova, Janssen, Medtronic, Merck, Occlutech, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Roche, Sequana, SQ Innovation, and Vifor; Dr Ferreira has received research support from Boehringer-Ingelheim, Novartis and

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Data Availability

Researchers should use the <https://vivli.org/link> to request access to study data and visit <https://www.mystudywindow.com/msw/datasharing> for further information.

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Ethical Approval

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